STANDARD OPENS OF THE CHART COME FROM STANDARD OPENS OF PROJ

THE STACKS PROJECT AUTHORS

Allowed standard prerequisites (editorial).

- Topology of Proj. algebra-lemma-topology-proj (cross-chapter, named). The homeomorphism D_+(f) = Spec(S_(f)) through which the correspondence is read.

Source reference. Original reference 00JP: focused-seeds/geometry/S9-cons-standard-open-proj/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a graded ring. Let $f \in S$ be homogeneous of positive degree. Suppose that $D(g) \subset \mathrm{Spec}(S_{(f)})$ is a standard open. Then there exists an $h \in S$ homogeneous of positive degree such that $D(g)$ corresponds to $D_+(h) \subset D_+(f)$ via the homeomorphism of Algebra, Lemma 00JP. In fact we can take h such that $g = h/f^n$ for some n.*

Proof. Write $g = h/f^n$ for some h homogeneous of positive degree and some $n \geq 1$. If $D_+(h)$ is not contained in $D_+(f)$ then we replace h by hf and n by $n+1$. Then h has the required shape and $D_+(h) \subset D_+(f)$ corresponds to $D(g) \subset \mathrm{Spec}(S_{(f)})$. $\square$

Date: stacks-project:constructions.tex:local-snapshot-20261012; GNU FDL 1.2 or later.